Finite Element Analysis of Cantilever H-Beam and I-Beam Structures in Bending, Torsion, and Transient Vibration

Camilo Garrido Fernandez
California Institute of Technology

This study presents a finite element analysis of cantilevered aluminum beam structures using ANSYS Mechanical. The investigation consists of three complementary analyses: a modal analysis of a 6063-T5 aluminum H-beam, a static structural analysis of a 6061-T6 aluminum I-beam subjected to combined bending and torsion, and a transient structural analysis examining the dynamic response of the same I-beam following load removal.

The modal analysis identifies the natural frequencies and mode shapes of the cantilever beam and evaluates how these frequencies change when additional mass is added to the free end. Static simulations are then used to compare finite element predictions of displacement, twist angle, and stress with analytical beam theory. Finally, transient simulations are used to study the free vibration response of the structure, from which decay rates, oscillation frequencies, and damping ratios are extracted using discrete Fourier transform analysis.

Results demonstrate strong agreement between analytical beam theory and FEA predictions for both bending and torsional deformation. Modal analysis of the H-beam revealed a fundamental weak-axis bending frequency of 6.042 Hz and a torsional mode at 49.821 Hz; the addition of a 3-foot cross-bar was found to reduce the torsional frequency by approximately 96%, shifting it to 2.0699 Hz. In transient simulations, the dominant frequencies extracted from stress time-series closely matched the theoretically predicted modal frequencies of 96.347 **Hz and** 160.530 **Hz. These findings demonstrate the significant impact of polar mass moment of inertia on structural vibration characteristics and validate the use of FEA to characterize how structures vibrate when loaded and unloaded.**

Nomenclature

b_A, b_B	Beam width and load arm length	m
E	Young's modulus	Pa
F	Applied force	N
f	Frequency	Hz

G	Shear modulus	Pa
h	Beam height	m
I	Second moment of area	m^4
J	Torsion constant	m^4
L	Beam length	m
M	Bending moment	$N \cdot m$
T	Torque	$N \cdot m$
t_A, t_B	Wall and end cap thickness	m
V	Shear force	N
α	Exponential decay rate	$1/s$
δ	Displacement	m
Δt	Simulation time step	s
ζ	Damping ratio	–
ν	Poisson's ratio	–
ρ	Density	kg/m^3
σ	Normal stress	Pa
τ	Shear stress	Pa
τ_c	Time constant	s
ϕ	Twist angle	rad
ω	Angular oscillation frequency	rad/s

I. Introduction

Beam structures are fundamental structural elements used throughout mechanical and aerospace engineering. Their mechanical behavior under bending, torsion, and dynamic loading conditions is well described by classical beam theory, yet complex geometries and loading configurations frequently require numerical methods such as finite element analysis (FEA) for accurate prediction of structural response.

This report investigates the structural behavior of aluminum beam structures through a sequence of three finite element analyses performed using ANSYS Mechanical [1]. The first portion of the study analyzes the modal characteristics of a

cantilevered architectural H-beam. Modal analysis provides insight into the natural frequencies and deformation shapes of a structure, which determine how the system responds to dynamic excitation.

The second portion of the study focuses on a static structural analysis of a rigid I-beam subjected to combined bending and torsional loading. Analytical beam theory is used to estimate expected displacements and twist angles, which are then compared with numerical simulation results.

Finally, a transient structural simulation is performed to analyze the dynamic response of the I-beam after the applied loads are removed. The resulting time-domain stress signals are analyzed using discrete Fourier transforms to extract oscillation frequencies and damping properties of the system.

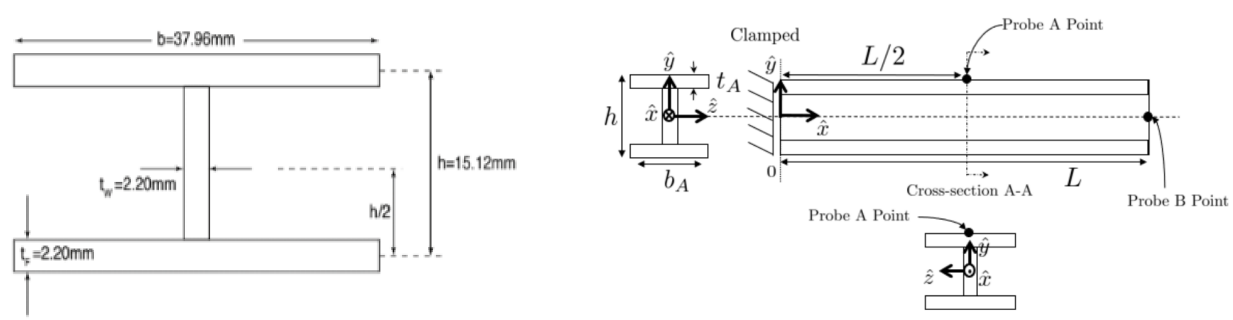
Together, these analyses provide a comprehensive study of beam mechanics spanning modal analysis, static deformation, and transient structural dynamics.

II. Methodology

The investigation was conducted in three distinct numerical phases using ANSYS Mechanical. This section details the geometric modeling, material properties, meshing strategies, and the derivation of boundary conditions for the modal, static, and transient analyses.

A. H-Beam Modal Analysis Setup

The first phase of the investigation involved determining the natural frequencies and mode shapes of a 6-foot architectural 6063-T5 aluminum H-beam — specifically, McMaster-Carr #4558T55 [2]. The geometry was imported as a STEP file and simplified by removing external fillets. The cross-sectional definition used for the finite element model is detailed in Figure 1. The material properties for 6063-T5 aluminum used in the simulation were pulled from MatWeb [3].



(a) Nominal dimensions. **(b) Detailed H-Beam FEA schematic.**

Fig. 1 Cross-sectional geometry of the 6063-T5 aluminum H-beam specimen.

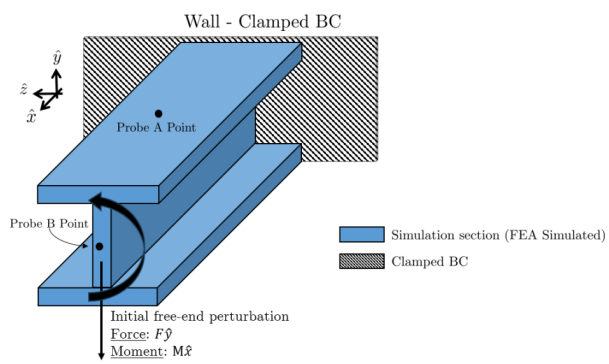
The beam was modeled as a cantilever with a fixed boundary condition applied to the face contacting the wall. No external loads were applied. An ANSYS modal analysis was conducted to determine the natural vibration characteristics

of the structure.

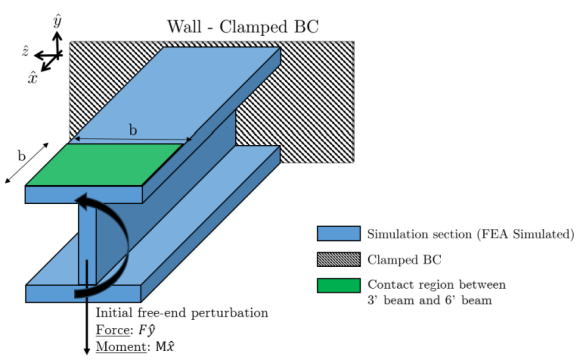
The finite element mesh was generated using a controlled sweep method to ensure high-quality quadrilateral elements. Adaptive sizing was disabled in favor of a proximity-based refinement strategy, with the *Num Cells Across Gap* parameter set to 2 to resolve the H-Beam wall thickness t_A.

The discretization along the longitudinal axis was defined by a *Sweep* method applied to the H-Beam body, using the cross-section coincident with the origin as the manual source. A mapped surface mesh was enforced on this source face using a *Face Meshing* constraint with an *All Quad* free face mesh type. Finally, the longitudinal resolution was explicitly defined by setting the *Number of Divisions* to 125, resulting in a structured mesh with uniform element distribution along the beam length L.

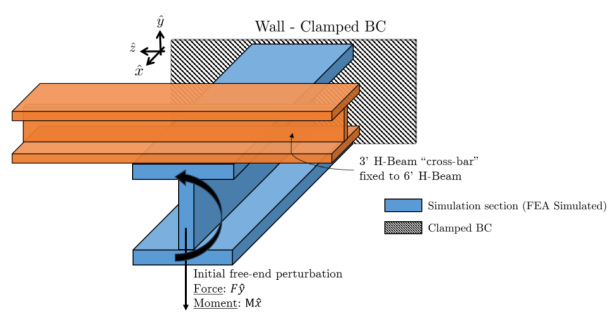
To evaluate the sensitivity of modal frequencies to end-mass conditions, a secondary assembly was modeled featuring a 3-foot cross-bar symmetrically clamped at the top of the free end of the first H-beam (Figure 2). This configuration was used to investigate the reduction in torsional frequencies due to increased polar mass moment of inertia.



(a) Cantilever H-beam (baseline).



(b) Cantilever H-beam and clamping interface.



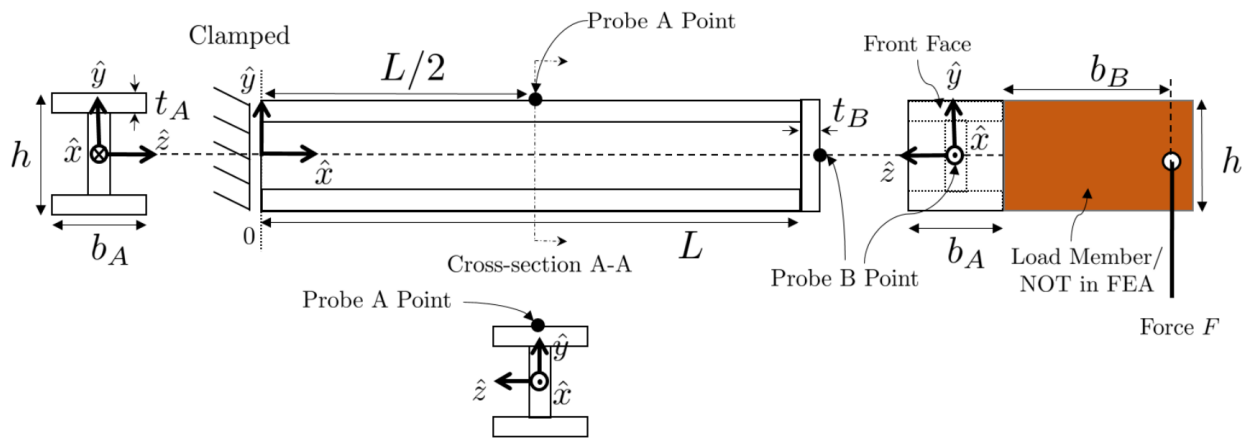
(c) Cantilever H-beam with clamped cross-bar.

Fig. 2 Isometric views of the H-beam simulation geometries for modal analyses.

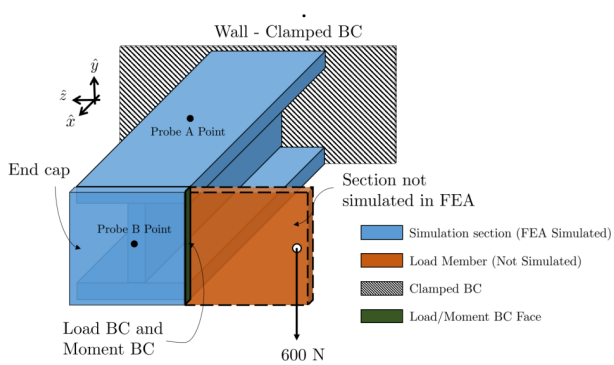
B. I-Beam Static and Transient Setup

For the static and transient analyses, a more rigid 6061-T6 aluminum I-beam was utilized. This setup involved a rigid load member attached to an end-cap on the I-beam's free face. This allows for the application of eccentric loads, inducing combined bending and torsion (Figure 3).

The geometry is detailed in Figure 3 while the specific material, geometric, and loading parameters used for this phase of the simulation are summarized in Table 2.



(a) Detailed schematic of the geometry of the assembled I-beam structure and the location of the applied load. All thicknesses in the cross-section of the I-beam are equal (t_A). The actual dimensions are shown in in Table 2



(b) 3D sketch of the I-beam geometry, end-cap, and load member assembly with boundary conditions.

Fig. 3 Geometry, assembly, and loading diagrams for the static structural and transient dynamic simulations.

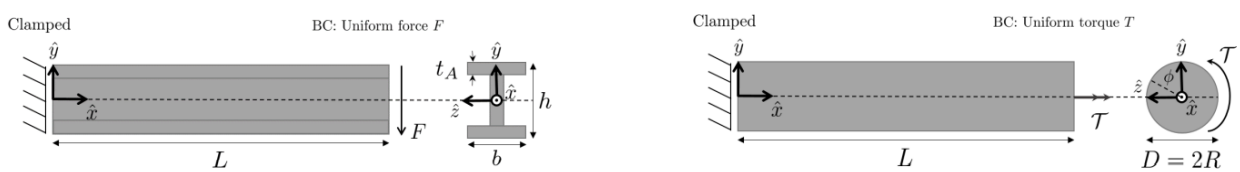
The mesh was generated using a *Sweep* method with quadrilateral elements along the length of the beam (*Number of Divisions* was left as "Default"). A second *Sweep* method was applied along the width of the beam, with *Number of Divisions* set to 20. To ensure converged results, two elements were placed through the wall thickness of all flanges and webs, following the procedure described for meshing the H-beam.

Boundary conditions for the simulation section were derived via a free-body diagram (FBD) analysis of the load member. The applied tip force F was translated into an equivalent force-moment system applied to the front face of

Table 2 Material, geometric, and loading parameters for the I-beam simulation assembly. The material properties listed were gathered from MatWeb.[4]

Category	Parameter	Value (Imperial)	Value (SI)
Material	E (Young's modulus)	10000 ksi	68.9 GPa
	G (Shear modulus)	3770 ksi	26.0 GPa
	ν (Poisson's ratio)	0.33	0.33
	ρ (Density)	0.0975 lb/in^3	2700 kg/m^3
Geometry	L (Beam length)	3.000 ft	0.9144 m
	h (Height)	3.000 in	0.0762 m
	b_A (I-Beam Width)	2.509 in	0.0637 m
	t_A (Web/flange thickness)	0.349 in	0.00886 m
	t_B (End cap thickness)	0.125 in	0.00318 m
	b_B (Load arm)	3.937 in	0.1 m
Loading	F (Force magnitude)	134.86 lbf	600 N

the simulation end-cap. Numerical results for tip displacement and twist angle were verified against first-principles estimates based on classical analytical models for cantilevered bending and torsion (Figure 4).



(a) Euler-Bernoulli I-beam bending diagram. **(b) Saint-Venant circular beam torsion diagram.**

Fig. 4 Classic beam loading diagrams used to determine simulation boundary conditions and verify FEA results.

C. Transient Analysis Parameters

For the transient structural analysis, the time step Δt was strictly controlled by the Nyquist criterion:

$$\Delta t \leq \frac{1}{2 f_{max}} \tag{1}$$

where f_{max} corresponds to the highest expected modal frequency (the torsional mode). The load was applied as a ramp over t_1 and instantaneously removed at t_2 to initiate free vibration. The resulting stress time-history data was then processed using a Fast Fourier Transform (FFT) to extract the system's dynamic properties.

The loading profile followed a stepped-ramp function as illustrated in Figure 5. The load was held constant at its maximum value for a stabilization period, and then rapidly released to initiate free vibration. The damping characteristics were extracted over the course of this free vibration phase.

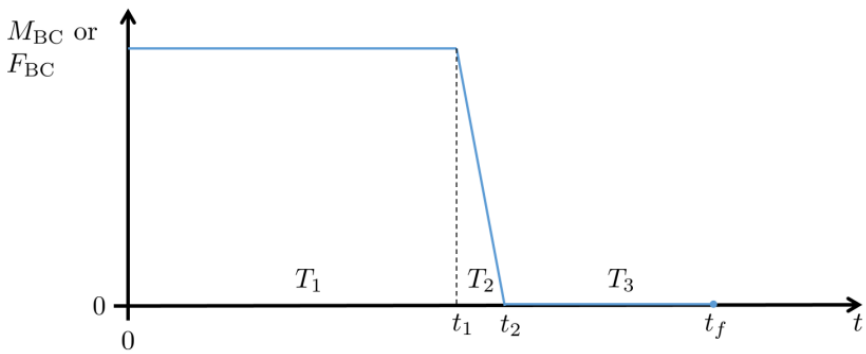


Fig. 5 Transient boundary condition profile showing the load hold, unloading, and free vibration phases.

The resulting normal (σ_x) and shear stress (τ_{XZ}) time-history data at probe A is then processed using a MATLAB [5] script that performs a regression-based decomposition to calculate the exponential decay rates α_1 and α_2 (with units of s$^-$1), and the oscillating frequencies ω_1 and ω_2 (with units of rad/s) corresponding to:

$$\sigma_x(t) = \bar{\sigma}_x e^{-\alpha_1 t} \cos(\omega_1 t + \psi_1) \tag{2}$$

and

$$\tau_{XZ}(t) = \bar{\tau}_{xz} e^{-\alpha_2 t} \cos(\omega_2 t + \psi_2) \tag{3}$$

where t is time in seconds, and $\bar{\sigma}_x$, $\bar{\tau}_{xz}$, and ψ_1, ψ_2 represent amplitude and phase constants (not calculated). The exponential decay rates are obtained using a least-squares fit to the waveform envelope, while the oscillation frequencies are computed using a discrete Fourier transform (DFT) applied to each time series — both are executed by the MATLAB script.

III. Results

A. Modal Analysis of Aluminum 6063-T5 H-Beam Structures

The first ten eigenmodes were extracted from the simulation. Table 3 summarizes the first four modes and their corresponding deformation types.

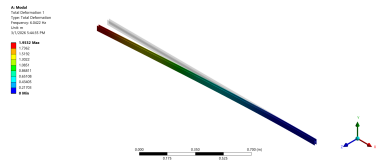
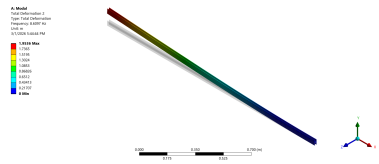
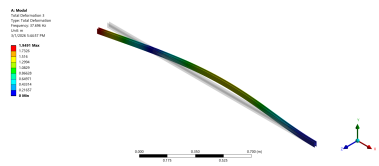
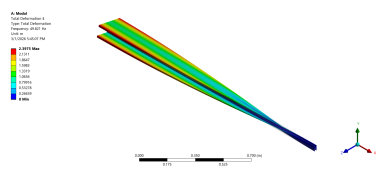
1. Classification of the First Four Modes

(i) **Mode 1: First-Order Weak-Axis Bending (6.042 Hz)**

This mode is characterized by lateral displacement primarily along the weak bending axis of the H-beam cross-section. The beam exhibits a smooth curvature with no internal stationary points along its span. The cross-section translates without significant distortion or torsional rotation.

(ii) **Mode 2: First-Order Strong-Axis Bending (8.610 Hz)**

Table 3 First 4 vibration modes of the selected 6' 6063-T5 aluminum alloy H-beam.

Mode Number	Frequency (Hz)	Dominant Deformation Type	Mode Shape Image	Mode Shape Description
1	6.042	First-Order Weak-Axis Bending		Simple cantilever sway about the weak-axis
2	8.610	First-Order Strong-Axis Bending		Simple cantilever sway about the strong-axis
3	37.696	Second-Order Weak-Axis Bending		Curvature about weak-axis, with a sign-change that leads to a second stationary point (away from fixed end)
4	49.821	First-Order Torsion		Simple twist about the long axis, with increasing stretching along $\hat{x}$ and $\hat{y}$ toward the free end

The beam deflects along the strong bending axis of the cross-section. The displacement profile shows a single smooth curvature with no nodes along the beam length. The increased frequency relative to Mode 1 reflects the higher geometric stiffness of the strong axis.

(iii) **Mode 3: Second-Order Weak-Axis Bending (37.696 Hz)**

This involves higher-order bending about the weak axis. The displacement profile contains an inflection point along the span where the curvature changes sign, producing a second stationary point. This harmonic behavior is typical of second-order bending.

(iv) **Mode 4: First-Order Torsion (49.821 Hz)**

This mode is dominated by torsional deformation about the longitudinal axis. The cross-section rotates about the beam axis, with opposite edges moving in opposing vertical directions, indicating twisting with minimal bending translation.

Note: the axes in the ANSYS contour plots shown in Table 3 are not aligned with the axes in Figure 1b and should be disregarded. In the discussion that follows, all axes and quantitaive results mentioned will be in reference to the coordinate frame in 1b and not the coordinate frame of the ANSYS results. Contour plots should be understood relative

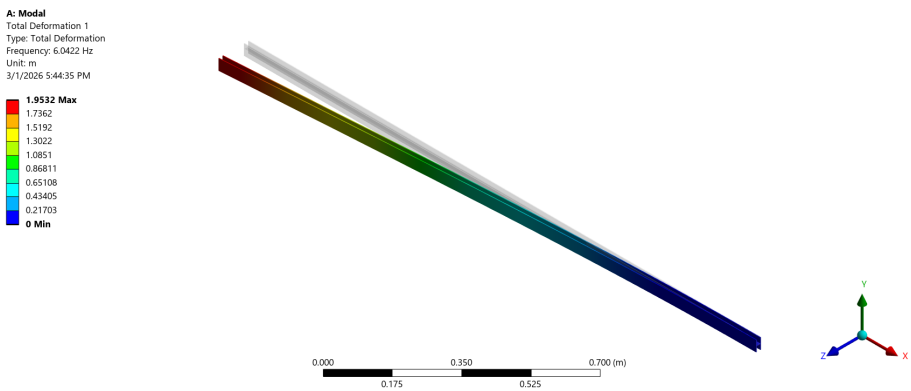
to the weak and strong axes of the H-beam.

2. Modes Excited by an Applied Perturbation

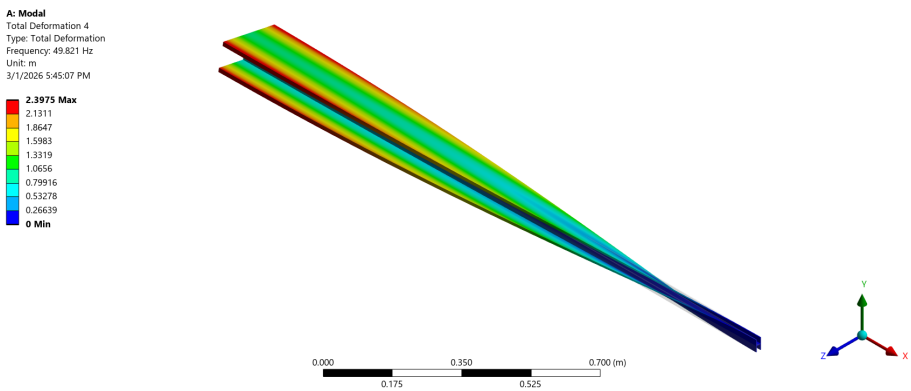
If a system is perturbed through some initial deformation, only modes sharing compatible symmetry with this deformation will be excited. To build intuition, consider a system that is allowed to vibrate freely, with force and bending boundary conditions set to zero at one end. Suppose a perturbation consisting of both twisting with a torque directed about $\hat{x}$, and bending caused by a force directed in $-\hat{y}$, were applied at the beam's free end. The applied force in the $-\hat{y}$ direction produces global bending displacement along the weak axis of the beam. The resulting deformation is smooth and monotonic along the span, which matches the spatial symmetry of the first-order weak-axis bending mode. Therefore, the dominant excited bending vibration is mode 1 — first-order weak-axis bending — with a frequency of $f = 6.042$ Hz. Higher-order weak-axis bending modes are not strongly excited because the imposed deformation is applied entirely at the free end. It lacks the internal node required by second-order bending shapes.

The applied moment about the $\hat{x}$ axis produces a uniform twisting deformation about the beam's long axis. This deformation is consistent with the spatial symmetry of the first-order torsional mode, in which the cross-section rotates about the long axis of the beam without lateral translation. Thus, the torsional perturbation excites mode 4 — first-order torsion — with a frequency of $f = 49.821$ Hz.

The mode shapes of these excited modes plotted in Figure 6.



(a) Mode 1: First-Order Weak-Axis Bending, f = 6.042 Hz.



(b) Mode 4: First-Order Torsion, f = 49.821 Hz.

Fig. 6 Excited mode shapes of H-beam based on an intitial perturbation consisting of a torque directed about $\hat{x}$ and a force directed in $-\hat{y}$, both applied at the end of the beam.

3. Maximum Time Step from Nyquist Criterion

To accurately capture the highest frequency mode in a discrete time simulation, the sampling time step must be small enough for at least two consecutive measurements to be on the same "wave" of its oscillation. This is described by the Nyquist criterion:

$$\Delta t_{max} = \frac{1}{2 f_{max}} \tag{4}$$

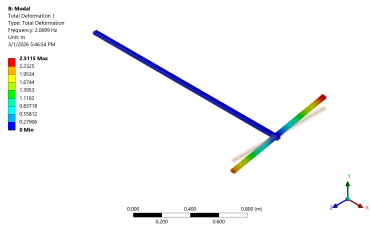
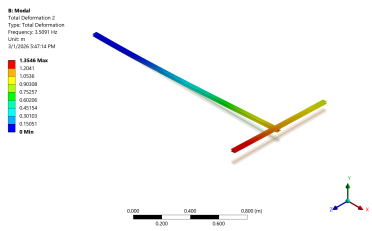
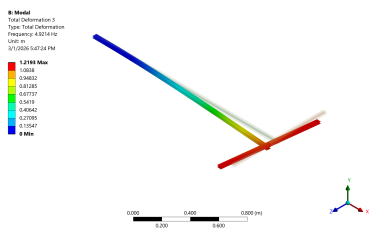
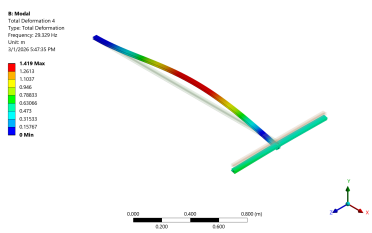
Inputting the highest expected modal frequency (the torsional mode), f_{max} = 49.821 Hz, into equation (4) yields the longest allowable time step: $\mathbf{\Delta t_{max} \approx 0.010}$ **s**.

4. Effect of Adding a Cross-Bar

Adding the 3 ft cross-bar increases both the mass and polar moment of inertia of the system. As a result:

- Bending frequencies decrease due to increased effective mass.
- Torsional frequency decreases significantly due to increased rotational inertia.

Table 4 First 4 vibration modes of the selected 6' 6063-T5 aluminum alloy H-beam with a 3' cross-bar symmetrically clamped at the free end.

Mode Number	Frequency (Hz)	Dominant Deformation Type	Mode Shape Image	Mode Shape Description
1	2.0699	First-Order Torsional Bending		The structure twists about the long axis of the base H-beam
2	3.5091	First-Order Weak-Axis Bending		Simple cantilever sway about the weak-axis of the base H-beam
3	4.9214	First-Order Strong-Axis Bending		Simple cantilever sway about the strong-axis of the base H-beam
4	29.329	Second-Order Weak-Axis Bending		Curvature about weak-axis of the base H-beam, with a sign-change that leads to a second stationary point

The modal frequencies obtained for the modified structure are summarized in Table 4.

Now suppose that a perturbation consisting of both twisting about $\hat{x}$ and bending about $\hat{y}$ is applied at the free end of this structure, exciting the first-order bending (about the weak-axis of the H-beam) and first-order torsion (about the long-axis of the H-beam) modes. The frequency of the first-order bending mode is now the larger of the two, at $f = 3.5091$Hz. Inputting this into equation (4) — the Nyquist criterion — yields the longest allowable time step to capture the expected deformation frequency response: $\mathbf{\Delta t_{max} \approx 0.285\ s}$. Because the frequencies of the modes we expect to be excited are now lower, they can be captured with larger time steps.

Interestingly, the frequency of the torsion mode decreased by almost 96% compared to the original geometry, becoming comparable (and in fact slightly lower) in magnitude to the bending modes. This demonstrates how large an

impact a significant change in polar mass moment of inertia — such as the addition of the cross-bar — can have on the natural torsion frequency of a structure, even if it does not dramatically change the torsional stiffness of the main beam.

B. Static Structural Analysis of Aluminum 6061-T6 I-Beam

1. Concept Review

For a cantilever beam subjected to a tip force F, the shear-force and bending-moment distributions are

$$V(x) = F \tag{5}$$

$$M(x) = \int V(x)dx \tag{6}$$

From the flexure formula, the bending stress at the top surface of the beam is

$$\sigma_x = \frac{M(x)y}{I} \tag{7}$$

The vertical displacement of the cantilevered beam tip along $-\hat{y}$ is given by

$$\delta_{x=L}(F) = \frac{FL^3}{3EI} \tag{8}$$

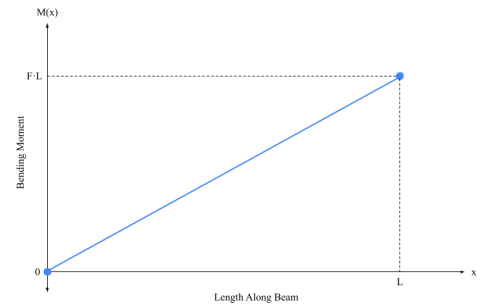
Noting that for a symmetric I-beam like the one in Figure 3a

$$I = \frac{1}{12}t_A\left(h - 2t_A\right)^3 + 2\left(\frac{1}{12}b_A t_A^3 + b_A t_A\left(\frac{h - t_A}{2}\right)^2\right) \tag{9}$$

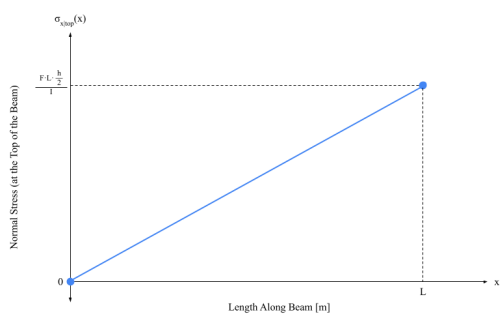
Using these relations, the theoretical distributions for the shear force and bending moment distributions for the I-beam setup in Figure 4a are shown in Figure 7.

Note: in equations (5)-(8), as well as the diagrams in Figure 7, the quantity F is treated as the magnitude of a force applied at the tip of the cantilever beam, acting in the $-\hat{y}$ direction, following the setup shown in Figure 4a. Furthermore, the internal bending moment $M(x)$ is defined as positive when it induces tension on the top surface of the beam at $y = +\frac{h}{2}$ (i.e., when the beam is concave downward in bending).

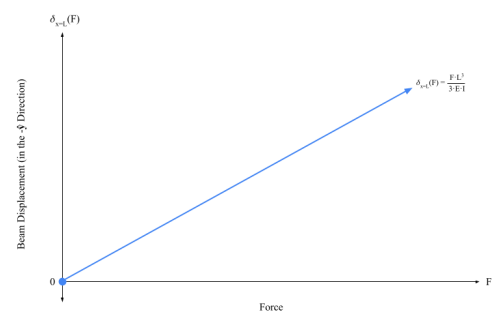
(a) Shear force distribution $V(x)$.



(b) Bending moment distribution $M(x)$.



(c) Normal stress $\sigma_{x|top}(x)$ at $y = h/2$.



(d) Tip displacement $\delta_{x=L}$ vs. Force F.

Fig. 7 Theoretical distributions and load-displacement relationship for the cantilever I-beam in bending.

2. Torsion Relations

For a cylindrical beam subjected to a torque, T, as in Figure 4b, the internal torque is constant (T) along the span, and zero at the ends. the twist angle is

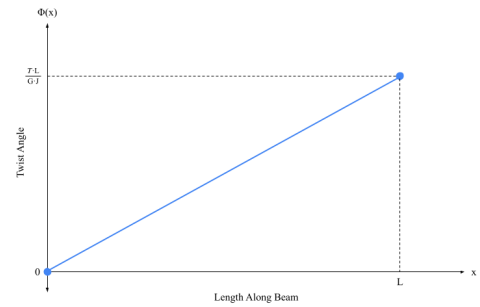
$$\phi(x) = \frac{Tx}{GJ} \tag{10}$$

The shear stress at $r = R$, $\tau|_R$, is given by

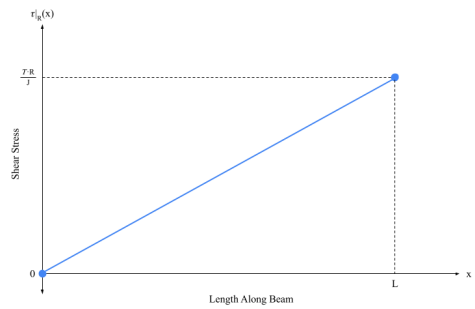
$$\tau|_R(x) = \frac{TRx}{JL} \tag{11}$$

These relations are plotted in Figure 8.

(a) Torque $T(x)$.



(b) Twist angle $\phi(x)$.



(c) Shear stress $\tau|_R(x)$.

Fig. 8 Theoretical distributions for the cantilever in torsion.

For a symmetric thin-walled I-beam cross-section, the torsion constant J can be approximated using

$$J \approx \frac{1}{3}(h - 2t_A)t_A^3 + \frac{2}{3}bt_A^3 \tag{12}$$

Assuming warping is unconstrained (consistent with Saint-Venant torsion theory), the maximum shear stress τ_{max} in a thin-walled open cross-section occurs at the surface of the thickest rectangular segment. For this uniform I-beam, it is expressed as:

$$\tau_{max} \approx \frac{Tt_A}{J} \tag{13}$$

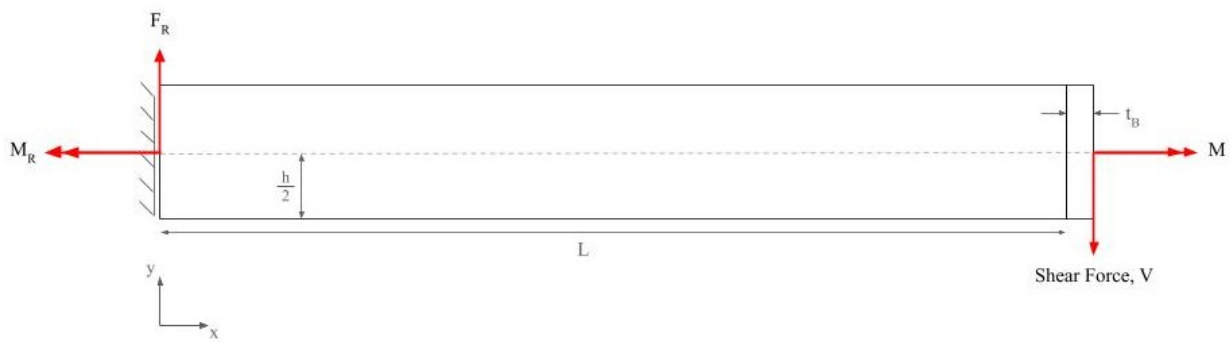
3. Mathematical Tools

The geometric and material parameters discussed above serve as the inputs for the numerical solver. In the context of the FEA modal and transient simulations, the structural response is governed by the characteristic equation of the system. To interpret these results, we recall that an *eigenvector* of a linear operator is a non-zero vector that retains its direction during transformation, while an *eigenvalue* (λ_i) represents the scalar factor by which that vector is scaled. Physically, in structural mechanics, these eigenvectors correspond to the mode shapes of the beam, and the eigenvalues relate directly to the natural frequencies (ω_n) at which the structure stores and releases elastic energy.

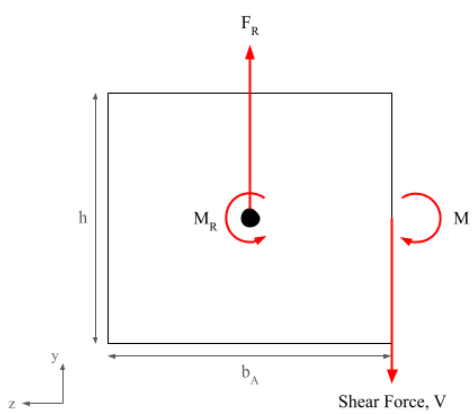
While static analysis provides a snapshot of peak stress and deformation, the transient structural simulation generates time-series data that captures the beam's evolution after load removal. To extract the underlying physical properties from these complex waveforms, we utilize the *Fourier Transform*. This mathematical operation decomposes a continuous time-domain signal into its constituent frequencies, translating the response into the frequency domain to identify dominant modes of vibration. For the digital signal processing required here, the *Discrete Fourier Transform* (DFT) is employed via the Fast Fourier Transform (FFT) algorithm — the most commonly used algorithm for computing the DFT — to calculate the oscillation frequencies (ω_d) and decay characteristics of the beam.

4. First Principles Estimates

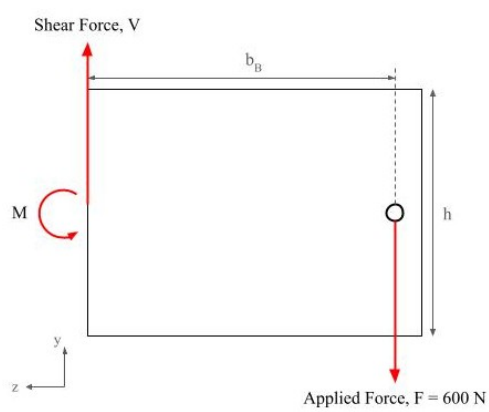
The boundary conditions implemented in the finite element simulation for the section shown in Figure 3a were determined by first-principles force and moment balances of the load member and the simulated I-beam structure.



(a) Simulated I-beam structure FBD ($\hat{x}$-$\hat{y}$ plane).



(b) Simulated I-beam structure FBD ($\hat{y}$-$\hat{z}$ plane).



(c) Load member FBD.

Fig. 9 Free-body diagrams (FBD) used to determine the simulation boundary conditions.

Free-Body Diagrams

- **Load Member:** Subjected to the applied load F and the reaction forces/moments at the interface with the I-beam

structure shown in the simulation.

- **Simulation Section:** Subjected to the equal and opposite loads at the front face of the end cap, as well as the resulting reactions at the fixed face.

Through force and moment balance of the load member:

$$\sum F_y = 0 \implies F + V = 0 \implies V = -F \tag{14}$$

$$\sum M_x = 0 \implies -b_B F + M = 0 \implies M = b_B F \tag{15}$$

The simulation section then experiences equal and opposite loads at the shared face with the load member. Thus, the boundary conditions to be applied in the simulations are:

- **Force:** $F = -600\ \text{N}\ \hat{y}$.
- **Moment:** $M = -b_B F = -60\ \text{Nm}\ \hat{x}$

If the beam were loaded in bending only, with this calculated boundary condition force applied directly to the front face of the end cap, then (using equation (8)) the vertical displacement δ_y at the tip ($x = L$) would be $\delta_{x=L}(F = -600\ \text{N}) \approx -1.562 \times 10^{-3}$ m.

Assuming boundary conditions that allow for warping, the twist angle ϕ at the midpoint ($x = L/2$) for the beam loaded in torsion only is:

$$\phi(L/2) = \frac{T_{applied}(L/2)}{GJ} \tag{16}$$

Using the expression for the I-beam torsion constant, J, from Equation 12, and the I-beam dimensions in Table 2, one finds that $J \approx 4.996 \times 10^{-8}$ m^4. Then, substituting this torsion constant, the calculated moment boundary condition of -60 Nm, and the shear modulus of the aluminum 6061-T6 I-beam (also in Table 2), the approximate twist angle is found to be:

$$\phi(L/2) \approx -0.0212 \text{ radians.}$$

It is important to note that the Saint-Venant torsion theory neglects the effects of warping, meaning that the cross-sections are free to out-of-plane deformation, offering less resistance to the applied torque. If warping is constrained, internal bi-moments are generated which have the mathematical effect of increasing the torsional rigidity of the member, thereby reducing the resulting twist angle for the same applied load. Thus, if the boundary conditions constrained warping of the I-beam, one should expect the twist angle $\phi(L/2)$ to be smaller in magnitude than the calculated value of $\phi(L/2) = -0.0212$ radians.

The Static Structural simulation was first conducted using only the torsion boundary condition, without accounting for gravity. The deformation in $\hat{z}$ was measured at the probe A location to be $\Delta x|_{L/2} = -6.1611 \times 10^{-4}$ m. Assuming

$\phi \ll 1$, we can use the small angle approximation $\tan\theta \approx \theta$:

$$\phi(L/2) \approx \tan\left(\frac{\Delta x|_{L/2}}{h/2}\right) \approx \frac{\Delta x|_{L/2}}{h/2} \approx -0.0162 \text{ radians}$$

This simulated value represents a roughly 23.6% reduction in magnitude compared to the first-principles estimate of -0.0212 radians. This deviation is primarily due to the fixed support boundary condition in the FEA model, which constrains the warping of the cross-section at the base. As previously noted, constraining warping introduces internal bi-moments that increase the effective torsional rigidity of the I-beam, leading to a smaller twist angle than predicted by the unconstrained Saint-Venant torsion theory.

Then the Static Structural simulation was conducted using both the bending and torsion boundary conditions (again ignoring gravity). The resulting deformation field is shown in Figure 10.

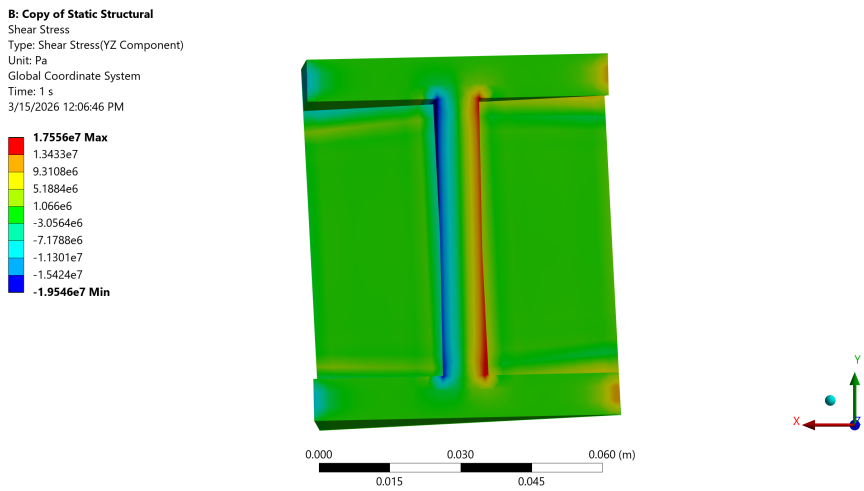
This time, the displacement at probe A measured $\Delta x|_{L/2} = -8.1236 \times 10^{-4}$ m. Approximating $\phi(L/2)$ as before,

$$\phi(L/2) \approx \frac{\Delta x|_{L/2}}{h/2} \approx -0.0213 \text{ radians}$$

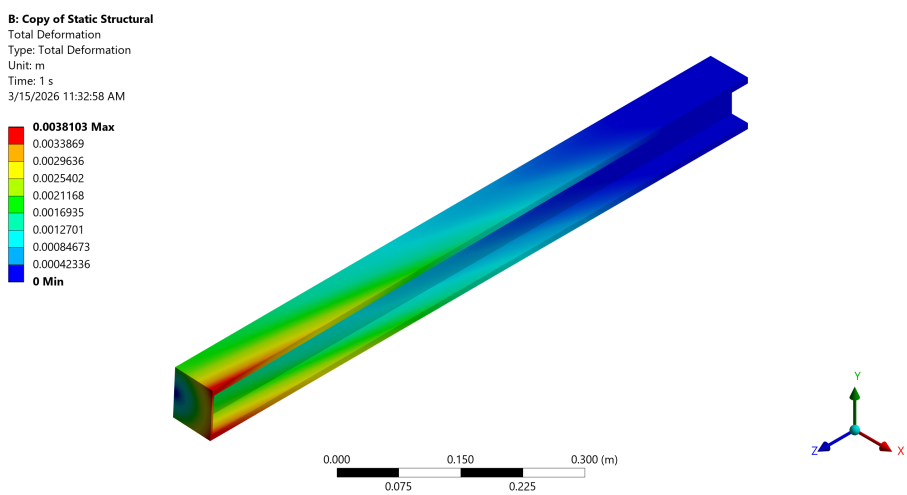
Comparing this to the value obtained in the pure-torsion simulation (-0.0162 radians), there is a noticeable increase in the calculated twist angle. This difference likely arises because the addition of the bending load introduces a large displacement in the $\hat{y}$ direction. Due to the asymmetric nature of the applied loads and the shifting of the reference neutral axis during bending, the purely lateral displacement approximation at probe A becomes coupled with the bending deflection, altering the pure torsion assumption and yielding a larger apparent twist angle.

The normal stress, σ_x, and the XZ shear stress, τ_{XZ}, were measured at probe A as 7.300×10^6 Pa and 1.750×10^7 Pa, respectively.

Additionally, the $\hat{y}$ displacement of the beam was recorded at probe B. It measured -1.5878×10^{-3} m, which is within a 1.660% error of the first-principles-based estimate of -1.562×10^{-3} m. This slight deviation is expected; analytical beam theories (such as Euler-Bernoulli) often neglect transverse shear deformation, making the analytical model artificially stiffer than the 3D solid model. The solid model captures these additional shear compliances, resulting in a slightly larger measured deflection.



(a) XZ shear stress, τ_{xz}, on cross-section A-A (Figure 3a).



(b) Total deformation contour.

Fig. 10 Static Structural results for the I-beam assembly under combined loading (no gravity).

Lastly, the Static Structural simulation was conducted using both the bending and torsion boundary conditions **and** standard earth gravity. The normal stress, σ_x, at probe A measured 7.426×10^6 Pa, and the $\hat{y}$ displacement at probe B measured -1.6286×10^{-3} m. These constitute changes of approximately 1.731% and 2.570%, respectively, from the measurements in the previous simulation conducted without accounting for gravity. Since these deviations are relatively minor, it would be appropriate to consider only the torsion and bending boundary conditions and ignore the gravitational load on the weight of the beam to simplify the simulation, as the results still easily fall within the targeted model accuracy of ±5%.

C. Transient Structural Analysis of Aluminum 6061-T6 I-Beam

1. Transient Time Interval Determination

To establish the temporal parameters for the transient simulation, the natural frequencies of the I-beam must first be considered. These frequencies, provided in Table 5, dictate the required duration for the load ramp and the necessary observation window for the subsequent free vibration.

Table 5 Tabulated frequencies for the first 3 vibration modes of the I-beam [6].

I-Beam Mode Type	Frequency (f)
Weak axis bending	50.715 Hz
Strong axis bending (f_{min})	96.347 Hz
Torsion (f_{max})	160.53 Hz

The simulation timing is divided into three phases: a stabilization hold phase, a rapid unload phase, and a free vibration phase. The ramp-up time, t_1, is designed to occur over 140 cycles of the fundamental strong-axis bending mode (f_{min} = 96.347 Hz) to minimize impulsive start-up transients. A hold time of 0.01 seconds is then added to establish the release time t_2. Finally, the total simulation time t_f is calculated to capture 16 full cycles of the bending response to ensure sufficient data for damping extraction. The resulting time intervals are calculated as follows:

$$t_1 = \frac{140}{f_{\min}} = \frac{140}{96.347 \text{ Hz}} \approx 1.4531 \text{ s} \tag{17}$$

$$t_2 = t_1 + 0.01 \text{ s} \approx 1.4631 \text{ s} \tag{18}$$

$$t_f = t_2 + \frac{16}{f_{\min}} = 1.4631 \text{ s} + \frac{16}{96.347 \text{ Hz}} \approx 1.6291 \text{ s} \tag{19}$$

2. Time Step Selection

The maximum allowable time step is again determined from the highest modal frequency via the Nyquist criterion. In this case, the torsional frequency, f_{max} = 160.53 Hz, is the highest. So,

$$\Delta t_{\max} \approx 3.11468 \times 10^{-3} \text{ s}$$

To improve resolution during the free vibration phase, the time step was made 10 times smaller:

$$\Delta t = \frac{\Delta t_{max}}{10} \approx 3.11468 \times 10^{-4}, \text{ during } T_3$$

Table 6 Extracted oscillation parameters during the free vibration phase.

Signal	Parameter	Value	Units
σ_x	α_1 (decay rate)	5.2512	s^{-1}
	f_1 (frequency)	96.5281	Hz
	ω_1 (angular frequency)	606.5040	rad/s
	ζ_1 (damping ratio)	0.0086578	–
τ_{XZ}	α_2 (decay rate)	13.7807	s^{-1}
	f_2 (frequency)	156.8582	Hz
	ω_2 (angular frequency)	985.5689	rad/s
	ζ_2 (damping ratio)	0.013981	–

3. Static Equilibrium Verification Befor Free Vibration

To verify that the transient simulation accurately captures the boundary conditions before free vibration begins, the system response was evaluated at $t = t_1$ and compared to the static structural results with combined torsion and bending boundary conditions (no gravity). At t_1, the applied loads have fully ramped up and are held constant, simulating a quasi-static state.

At probe A, the normal stress σ_x and shear stress τ_{XZ} at t_1 were evaluated to be 7.2700×10^6 Pa and 1.7494×10^7 Pa, respectively. These values closely match the pure static simulation results of 7.300×10^6 Pa and 1.750×10^7 Pa, exhibiting minimal deviations of approximately 0.41% and 0.03%, respectively. Because this transient response at t_1 demonstrates such a low margin of error compared to the static solid model, we can confidently conclude that the simulation has reached static equilibrium prior to load removal.

4. Stress Time Series Analysis

The stress signals were extracted at probe A and analyzed using MATLAB. The outputs of the MATLAB script are shown in Figure 11.

The extracted parameters are tabulated in Table 6.

The time constant is defined as:

$$\tau_c = \frac{1}{\alpha} \tag{20}$$

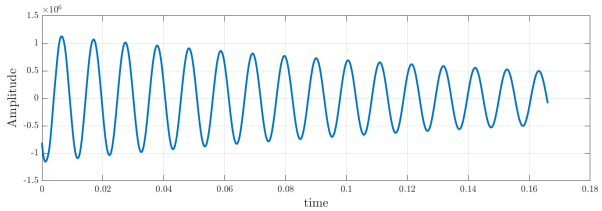
The damping ratio can be determined from the extracted decay rates α and oscillation frequencies ω in Table 6 in the following two ways:

1) Lightly damped approximation:

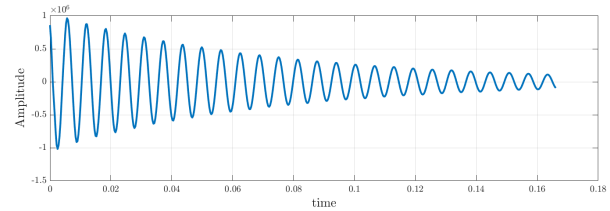
$$\zeta_{\text{approx}} = \frac{\alpha}{\omega} \tag{21}$$

2) Exact expression:

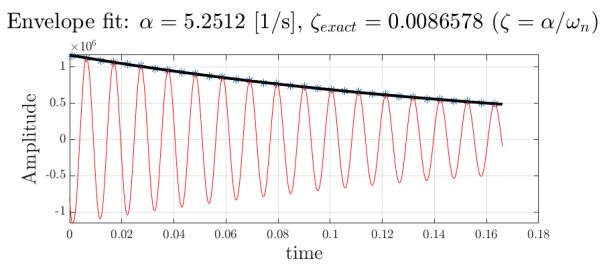
$$\zeta_{\text{exact}} = \frac{\alpha}{\sqrt{\alpha^2 + \omega^2}} \tag{22}$$



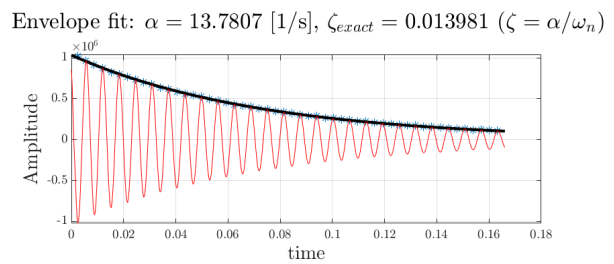
(a) σ_x during free vibration phase.



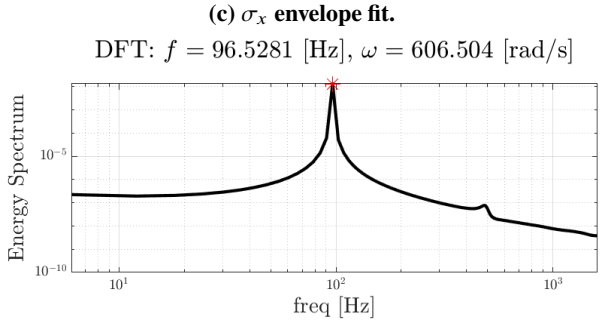
(b) τ_{XZ} during free vibration phase.



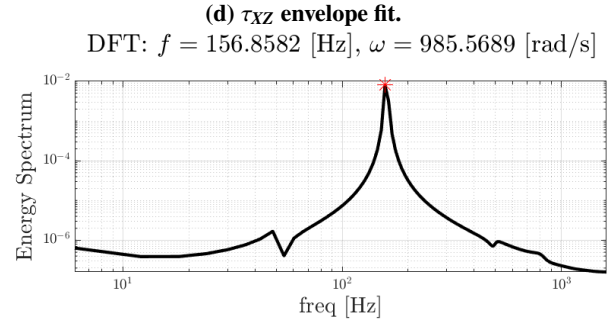
(c) σ_x envelope fit.



(d) τ_{XZ} envelope fit.



(e) σ_x energy spectrum.



(f) τ_{XZ} energy spectrum.

Fig. 11 Frequency and damping analysis of x-normal and xz-shear stresses during free vibration. The left column shows normal stress data and the right column shows shear stress data.

Substituting the oscillation parameters in Table 6 into equations (20)-(22) yields the results shown in Table 7.

Table 7 Calculated oscillation parameters during the free vibration phase.

Signal	Parameter	Value	Units
σ_x	$\tau_{\mathbf{c},\mathbf{1}}$ (time constant)	0.19043	s
	$\zeta_{\text{approx},\mathbf{1}}$ (approx. damping ratio)	0.0086581	–
	$\zeta_{\text{exact},\mathbf{1}}$ (exact damping ratio)	0.0086578	–
τ_{XZ}	$\tau_{\mathbf{c},\mathbf{2}}$ (time constant)	0.0725652543049	s
	$\zeta_{\text{approx},\mathbf{2}}$ (approx. damping ratio)	0.013982	–
	$\zeta_{\text{exact},\mathbf{2}}$ (exact damping ratio)	0.013981	–

As shown in Table 7, the values for ζ_{approx} and ζ_{exact} are nearly identical to each other, with differences appearing only in the sixth decimal place for both the normal and shear stress signals. Moreover, comparing these calculated values to the damping ratio reported by the ANSYS GUI confirms the accuracy of the extracted envelope fit, as ζ_{exact} perfectly matches the damping ration extracted from the envelope fit for both σ_x and τ_{XZ}.

Furthermore, because $\zeta \approx 0.0086 \ll 1$ for the bending mode and $\zeta \approx 0.0139 \ll 1$ for the torsion mode, the system is highly underdamped. This highlights the need for active control systems or dampers to prevent such sustained, disruptive oscillations when they are undesired.

Additionally, comparing the extracted mode frequencies f_1 and f_2 in Table 6 to the theoretical frequencies of the strong-axis bending and the torsion modes (respectively) in Table 5 confirms that these are indeed the two dominantly excited vibration modes, as predicted. The extracted frequency $f_1 = 96.5281$ Hz is within 0.188% error of the tabulated strong axis bending mode frequency of 96.347 Hz, while $f_2 = 156.8582$ Hz is within 2.29% error of the tabulated torsion mode frequency of 160.530 Hz

5. Calculation of Decay Periods

In structural and mechanical engineering, knowing the the settling time — how long a structure continues to oscillate after an excitation — is critical for everything from assessing the fatigue life of a part, designing a vehicle suspension system for passenger comfort, and achieving precise movements in robotics. A common benchmark for determining when a system has settled down is assessing when the amplitude decays to 1% of its initial peak. In this case, the number of full oscillation cycles (periods) N required for the stress magnitude to drop to 1% of its value at time t_2 can

be determined by examining the ratio of the response at time t relative to the start of the free vibration phase:

$$\frac{\sigma_x(t)}{\sigma_x(t_2)} = \frac{\bar{\sigma}_x e^{-\alpha_1 t}\cos(\omega_1 t + \psi_1)}{\bar{\sigma}_x e^{-\alpha_1 t_2}\cos(\omega_1 t_2 + \psi_1)} = 0.01 \tag{23}$$

The free vibration phase starts at time t_2, so the evaluation time, t, during this phase can be defined as N periods after t_2:

$$t = t_2 + \left(\frac{2\pi}{\omega}\right) N \tag{24}$$

Because the period $T = 2\pi/\omega$, the phase of the harmonic component at time t returns to its value at t_2:

$$\cos\left(\omega\left[t_2 + N\frac{2\pi}{\omega}\right] + \psi\right) = \cos(\omega t_2 + 2\pi N + \psi) = \cos(\omega t_2 + \psi) \tag{25}$$

Substituting this back into the ratio, the amplitude constants $\bar{\sigma}_x$ and the cosine terms cancel, leaving only the exponential decay of the envelope:

$$\frac{e^{-\alpha(t_2 + N\frac{2\pi}{\omega})}}{e^{-\alpha t_2}} = e^{-\alpha N\frac{2\pi}{\omega}} = 0.01 \tag{26}$$

Taking the natural logarithm of both sides:

$$-\alpha N\frac{2\pi}{\omega} = \ln(0.01) \tag{27}$$

Solving for N yields the final expression:

$$N = \frac{-\ln(0.01)\omega}{2\pi\alpha} \tag{28}$$

Finally plugging in the values for α_1 and ω_1, and α_2 and ω_2:

$$N_1 = \frac{-\ln(0.01)\omega_1}{2\pi\alpha_1} \approx 85$$

$$N_2 = \frac{-\ln(0.01)\omega_2}{2\pi\alpha_2} \approx 52$$

Meaning that it takes about 83 cycles for the amplitude of σ_x to settle down to within 1% of its peak, and 52 cycles for τ_{XZ} to do the same.

IV. Conclusion

This study successfully employed finite element analysis to characterize the modal, static, and transient behavior of aluminum H-beam and I-beam structures. Static results for bending and torsion demonstrated high correlation with Euler-Bernoulli and Saint-Venant theories, with transient verification at t_1 showing less than 0.5% deviation from static

equilibrium values. Modal analysis highlighted the critical role of mass distribution; specifically, the addition of a 3-foot cross-bar shifted the fundamental torsional mode from 49.821 Hz to 2.0699 Hz, a 96% reduction that underscores how distal mass affects structural bandwidth.

The transient analysis confirmed that dominant vibration frequencies correspond to the structural modes excited by the initial perturbation. The extraction of damping ratios ($\zeta \approx 0.01$) via FFT analysis confirmed that the structures are highly underdamped, necessitating careful consideration of settling times in precision engineering applications. These methodologies—combining high-fidelity FEA with first-principles verification—are directly applicable to the design of aerospace components, robotic manipulators, and architectural supports where dynamic stability and predictable load-path behavior are paramount.

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